

SymPy Bug Card

Bug 15: uniform quantile misses the $p = 1$ endpoint

Task. Compute the $p = 1$ quantile of a unit uniform random variable.

$$U \sim \text{Uniform}(0, 1), \quad Q(1) = \text{quantile}(U)(1).$$

SymPy's answer.

$$Q(1) = \emptyset.$$

Correct answer.

$$Q(1) = 1.$$

Explanation. The documented quantile definition is

$$Q(p) = \inf\{x \in \mathbb{R} : p \leq F(x)\},$$

where F is the cumulative distribution function. For $U \sim \text{Uniform}(0, 1)$, the CDF satisfies $F(x) < 1$ for $x < 1$ and $F(1) = 1$. Thus $Q(1) = 1$. SymPy's quantile API reports quantiles as sets, so the expected API-level result is $\{1\}$; the alarming behavior is that it returns $\emptyset$, meaning that no value is reported for this endpoint quantile.

Likely root cause. The generic continuous quantile implementation in `sympy/stats/crv.py` solves $F(x) = p$ over the support; for `Uniform(0, 1)` this produces an inverse intersected with `Interval.Ropen(0, 1)`, so substituting $p = 1$ drops the endpoint. Deduplication frames this as a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-015-uniform-quantile-misses-the-p-1-endpoint/](#)